

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

03

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P , Q , R , and S correspond to P' , Q' , R' , and S' , respectively. The measure of angle P is 30° , the measure of angle Q is 50° , and the measure of angle R is 70° . The length of each side of $P'Q'R'S'$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

- A. 10°
- B. 30°
- C. 40°
- D. 90°

The length of each side of a square is 94 centimeters (cm). Which expression gives the area, in cm^2 , of the square?

- A. $2 \cdot 94$
- B. $2 \cdot 94 \cdot 94$
- C. $4 \cdot 94$
- D. $94 \cdot 94$

A circle in the xy -plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

- A. The center is at $(13, k)$ and the radius is 8.
- B. The center is at $(k, 13)$ and the radius is 8.
- C. The center is at $(k, 13)$ and the radius is 64.
- D. The center is at $(13, k)$ and the radius is 64.

Triangle FGH is similar to triangle JKL , where angle F corresponds to angle J and angles G and K are right angles. If $\sin F = \frac{308}{317}$, what is the value of $\sin(J)$?

A. $\frac{75}{317}$

B. $\frac{308}{317}$

C. $\frac{317}{308}$

D. $\frac{317}{75}$

Q5

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In triangle JKL , the measures of $\angle K$ and $\angle L$ are each 48° . What is the measure of $\angle J$, in degrees? (Disregard the degree symbol when entering your answer.)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A right circular cylinder has a base diameter of 22 centimeters and a height of 6 centimeters. What is the volume, in cubic centimeters, of the cylinder?

- A. 132π
- B. 264π
- C. 726π
- D. $2,904\pi$

A circle in the xy -plane has its center at $-4, 5$ and the point $-8, 8$ lies on the circle. Which equation represents this circle?

A. $x - 4^2 + y + 5^2 = 5$

B. $x + 4^2 + y - 5^2 = 5$

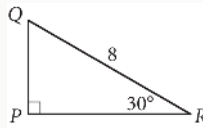
C. $x - 4^2 + y + 5^2 = 25$

D. $x + 4^2 + y - 5^2 = 25$

Q8

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In the right triangle shown above, what is the length of $\overline{PQ}$?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850. In triangle XYZ , the length of $\overline{YZ}$ is 60. What is the length of $\overline{XY}$?

- A. 204
- B. 182
- C. 60
- D. 12

A cube has an edge length of 41 inches. What is the volume, in cubic inches, of the cube?

- A. 164
- B. 1,681
- C. 10,086
- D. 68,921